

Poker Trainer — Complete Asset Creation Guide (Visual + Audio)

Everything you need to draw every image and source every sound, on an iPad + Apple Pencil, in one day, without needing to ask anything. Work top to bottom or jump around via the checklist at the end.

PART 0 — READ THIS FIRST: The two rules that keep you out of trouble

1. **TRANSPARENCY.** Almost every asset is a character, chip, icon, or card that sits ON TOP of a background. Those MUST be PNGs with a transparent background (alpha channel). The ONLY assets that are full-frame (no transparency needed) are the three room backgrounds. If you draw a character on a white square and export it, you'll get a white box around your dealer in-game. Always draw on a transparent canvas (no background layer).
2. **DON'T touch code or filenames casually.** The code already references exact filenames (e.g. `butler_profile.png`, `background_root.png`). If you name your file something else, it won't load. This guide gives you the EXACT filename for every asset. Name them exactly. When in doubt, hand the file to Claude Code and let it place it — see PART 6.

PART 1 — WHICH APP, AND HOW TO DRAW ON IT

Use Procreate. (Not Adobe Fresco/Photoshop on the iPad — here's why.)

- Procreate is a one-time ~\$13 purchase, runs great on old iPads, and is the standard for iPad illustration. It exports transparent PNGs trivially. If you already have it or can buy it, use it.
- If you can ONLY use Adobe: use **Adobe Fresco** (free, made for iPad + Pencil, exports transparent PNG). Photoshop-on-iPad works too but is heavier. Fresco is the closest Adobe equivalent to Procreate for drawing.
- Everything below is written for Procreate but the concepts (layers, transparent canvas, PNG export) are identical in Fresco.

Procreate crash course (enough to do everything today)

1. **New canvas:** Gallery → + (top right) → custom size. Set dimensions per asset (sizes in PART 2). Set DPI to 72 (screen art, not print). Color profile: sRGB.
2. **Transparent background:** Procreate canvases are transparent by default. To SEE the transparency, tap the Layers icon and hide/delete the "Background color" layer (tap it, toggle it off). The checkerboard = transparent = correct.
3. **Layers:** tap the two-squares icon. Draw different parts on different layers (e.g. dealer body on one, monocle on another) so you can fix one without redrawing everything. Keep the background-color layer OFF.
4. **Brushes:** tap the brush icon. For clean stylized art, the "Monoline" (Calligraphy set) or "Studio Pen" (Inking set) give crisp edges. For soft shading use "Soft Brush" (Airbrushing). You don't need many — an inking pen + a soft airbrush covers it.
5. **Color:** top-right circle. Tap-hold any color on canvas to eyedrop it.
6. **Undo:** two-finger tap. **Redo:** three-finger tap. **Erase:** the eraser icon (uses the same brushes).
7. **Selection/move:** the arrow (Transform) and the S-curve (Selection) tools to move, scale, reflip parts around.
8. **EXPORT (critical):** the wrench (Actions) → Share → **PNG** (NOT JPEG — JPEG has no transparency and will white-box your art). PNG preserves the alpha channel. Save to

Files or AirDrop to your computer.

The "draw it twice the size" rule

The spec says assets must render crisply up to 4K. Practical rule: **draw everything at 2x the size listed in PART 2**. If a chip renders ~120px in-game, draw it at 240px+. Bigger source art scaled DOWN looks crisp; small art scaled UP looks blurry. When unsure, draw bigger. Storage is cheap; pixelation is forever.

PART 2 — THE COMPLETE VISUAL ASSET LIST (draw all of these)

Filenames are EXACT — match them. All are transparent PNGs EXCEPT the three backgrounds.

A. BACKGROUNDS (3 files — full-frame, NO transparency, these are the room)

These are three blur variants of the SAME high-limit poker room. Draw the room ONCE, then make three versions at different blur levels.

- `background_root.png` — HEAVY blur (~12-20px). Root screen. Atmospheric mush; buttons sit on top.
- `background_mode.png` — MEDIUM blur (~6-10px). Mode Selection + Post-Round. Room shapes legible.
- `background_game.png` — LIGHT blur (~2-4px). Game screen. Room recognizable; table/dealer sharp on top. Aesthetic (from spec): a PRIVATE PREMIUM CARD ROOM. Dim lighting, leather + wood textures, brass fixtures, restraint. Explicitly NOT Vegas — no slot machines, no neon, no gambling-floor look. Think: a quiet, moody, expensive back room. Warm browns, deep shadows, a little brass glow.
- **How to make the 3 blurs:** draw the room sharp once (big — like 2560x1440+). Duplicate the layer/canvas 3 times. Apply Gaussian Blur (Adjustments → Gaussian Blur, slide with the pencil) at three strengths. Export each as its background name. Size: match a 16:9 canvas, e.g. 2560x1440 (bigger is fine).
- These are the heaviest files (~1.3MB combined target). Fine.

B. THE DEALER — BUTLER (3 files, transparent)

A stylized, illustrated (Persona-character idiom — stylized, not photorealistic) butler. Identity markers on ALL three: golden monocle, sharp defined mustache, visible eyebrows, and FACELESS otherwise — NO eyes, nose, or mouth. Butler-formal attire.

- `butler_profile.png` — SIDE profile, facing LEFT (toward the table). Dealing posture. Used on the Game screen.
- `butler_neutral.png` — FRONT-facing. Calm: both eyebrows level, monocle straight. Post-Round PASS state.
- `butler_raised.png` — FRONT-facing. The monocle-side eyebrow RAISED, that monocle slightly tilted, optional subtle forehead wrinkle. Post-Round FAIL state. Tips: draw the front-facing neutral first, then DUPLICATE it and just raise one eyebrow + tilt the monocle to make `butler_raised` (keeps them consistent). The profile is a separate drawing (side view). Draw large — he's a focal character. ~1200-1600px tall source.

C. THE DEALER — FROG easter egg (3 files, transparent)

Front-facing frog dealer (only ever front-facing). Wears the SAME golden monocle as the butler. Whimsical. The two expression files are OVERLAYS that sit on top of the base — so draw them positioned to line up over the base frog's face.

- `frog_base.png` — front-facing frog, no expression, with the golden monocle.

- `(frog_expression_pass.png)` — TRANSPARENT overlay: just two small light-pink blush circles, positioned to land on the frog's cheeks. Everything else transparent.
- `(frog_expression_fail.png)` — TRANSPARENT overlay: just the frog's tongue sticking out downward (small mocking expression). Everything else transparent. Tip: draw `frog_base`, then on a NEW layer draw the blush, hide the base, export just the blush layer = the pass overlay. Same for the tongue. They must align over the base when layered, so draw them in position relative to the base.

D. POKER TABLE (1 file, transparent-ish)

- `(table.png)` (name may differ — see PART 6 to confirm) — the poker table, rendered to sit INTEGRATED into the room with depth/lighting, NOT a flat sticker. Green/blue felt, wood or leather rail. It sits on top of `background_game.png` with the dealer to its right. Draw it at a size that fills the center table area; transparent around the table shape.

E. PLAYING CARDS (52 files, transparent)

52 individual card face PNGs. This is the most tedious set. Naming convention is usually rank+suit, e.g. `(card_AS.png)` (Ace of Spades), `(card_10H.png)` (Ten of Hearts), `(card_2C.png)`, `(card_KD.png)`, etc. — CONFIRM the exact naming with Claude Code before you start (PART 6), because 52 mis-named files is a bad day.

- **Massive time-saver:** do NOT draw 52 from scratch. Draw ONE card template (white rounded rectangle, border). Then for each card, place a big rank letter/number in a corner (and mirrored in the opposite corner, as real cards do) + the suit pip. Hearts/diamonds RED, spades/clubs BLACK. You can build all 52 by swapping the rank glyph and suit on the same template. Consider making the 4 suit pips once and reusing.
- **Even bigger time-saver:** honestly consider whether hand-drawing 52 cards is worth your day. Simple, clean, legible cards (rank + suit, classic look) are the goal — they must be READABLE at a glance more than artistic. If you want them uniform, you might build them in a way that reuses the template heavily. Prioritize legibility over flourish.
- Card size: real card proportions (~2.5:3.5). Draw ~500x700px each.

F. POKER CHIPS (8 files, transparent)

Eight denomination chips, identified by COLOR (per US cardroom convention). Filenames likely `(chip_white.png)`, `(chip_red.png)`, `(chip_green.png)`, `(chip_black.png)`, `(chip_purple.png)`, `(chip_yellow.png)`, `(chip_brown.png)`, `(chip_gold.png)` (confirm names — PART 6). Colors (standard, from spec): white, red, green, black, purple, yellow/orange, brown/grey, gold/blue — in ascending value order.

- Draw a top-down or slight-angle poker chip: a disc with the classic edge stripes (dashes around the rim in a contrasting color). Faces do NOT need large dollar numbers — at most a tiny inlay marking. Color IS the identifier.
- **Time-saver:** draw ONE chip, then recolor it 8 times (Procreate: use Hue/Saturation adjustment, or just fill layers). Keep the shape identical; only the color changes. They MUST be visually distinct from each other AND from the dealer button's blue/green (below).
- Chip size: ~200-300px each. They stack, so a clean simple disc reads best.

G. DEALER BUTTON (1 file, transparent)

- `(dealer_button.png)` — a circular disc, OCEAN BLUE base, chip-style edge with white rectangular inserts around the rim each containing a dice-face pattern; center has a stylized SAGE GREEN "M" monogram (the in-fiction venue mark, in lieu of "DEALER" text); an inset DASHED RING sits between the edge and the M. (That dashed ring doubles as the loading-screen progress arc, so keep it a clean circle.) Blue

+ green here are deliberately NOT chip colors so it never reads as a wagering chip.
~300px.

H. UI ICONS — persistent cluster + utility (transparent, small)

These are simple line/glyph icons, monochrome-ish (they get tinted by theme; draw them in a neutral light color like off-white on transparent). Filenames likely as listed; confirm PART 6.

- `icon_shop.png` — storefront
- `icon_help.png` — question mark "?"
- `icon_settings.png` — cog / gear
- `icon_home.png` — house
- `icon_close.png` — X (the Game-screen cluster's 4th icon)
- `icon_exit.png` — a DOOR glyph (Post-Round exit)
- `icon_copy.png` — two-papers copy glyph
- `icon_share.png` — square-with-up-arrow share glyph
- `icon_offline.png` — a cloud-with-slash (offline indicator)
- `icon_leaderboard.png` — a trophy (or similar) glyph
- `icon_tomato.png` — the tomato (in-app currency icon). This one can be full-color (a cute stylized tomato). Used in Shop/profile/leaderboard.
- `icon_sidepot.png` — small icon: three chips stacked vertically with a 4th leaning diagonally, all in distinct colors (shown in recap Overall row on side-pot hands).
- `icon_allin_marker.png` — an all-in marker for a seat (e.g., a colored ring or "ALL-IN" badge). Confirm if needed. Tips: keep these SIMPLE and readable at small sizes (they render ~30-50px). Bold, clean strokes. Draw at ~256px on transparent. Since theme code tints them, draw them light grey/white — don't bake in a color.

Anything else?

- **Root logo:** "POKER TRAINER" wordmark appears top-left on Root. If it's currently just text, you may not need an asset — but if you WANT a drawn logo, make `Logo.png` (transparent). Confirm whether the code expects a logo image or renders text (PART 6). Optional.

PART 3 — DRAWING TIPS FOR THE HARD/IMPORTANT ONES

- **The dealer is your hero asset.** Spend the most time here. Faceless-but-for-monocle-mustache-eyebrows is a strong, clean look — lean into silhouette and attire detail since he has no face to emote with. His "expression" is entirely eyebrow + monocle, so make those READ clearly (that's the whole pass/fail signal).
- **Keep a consistent light source** across all assets (e.g. light from upper-left). Dim, warm, single-source lighting sells the "private card room" mood.
- **Palette discipline:** warm browns, brass/gold, deep shadow, cream highlights. The chips and dealer-button are your only saturated pops of color. Everything else restrained.
- **Cards & chips = legibility first.** They're functional. A player must read them instantly. Clean > fancy.
- **Draw on transparent, always check the checkerboard** before exporting.
- **Name a layer per asset part** so fixes are easy.

PART 4 — THE COMPLETE AUDIO ASSET LIST (source all of these)

You need 9 SFX + music tracks. All SFX bundle into the app (<3MB total). Music is separate.

SFX (9 files — short, 1-3 seconds each)

Likely folder: `assets/audio/sfx/` with names like these (confirm exact names — PART 6):

1. `card_deal` — a single card-dealing/shuffle-riffle sound (plays once per hand, not per card).
2. `button_click` — a subtle, fast click/confirm (plays when focusing a math input).
Short + soft.
3. `chip_push` — chips sliding/clacking forward (Caller bet). A chip-shove sound.
4. `side_pot_split` — chips splitting into two piles (all-in side pot). A bigger chip sound.
5. `frog_toggle` — a small "bloop"/amphibian sound (easter-egg trigger). Whimsical, distinct.
6. `modal_swoosh_open` — a soft swoosh/whoosh (modal opens).
7. `modal_swoosh_close` — the complementary close swoosh (modal closes).
8. `slide_in` — the Game→Post-Round slide sound.
9. `slide_out` — the Post-Round→Game slide sound. `slide_in` + `slide_out` should feel like a matched PAIR (one can literally be the other reversed).

Where to get SFX (all free / royalty-free)

- **Freesound.org** — huge library, great for chip/card/click sounds. Filter by CC0 license to avoid attribution hassle. Search: "poker chips", "card deal", "card shuffle", "click ui", "swoosh", "whoosh", "bloop".
- **Pixabay (Sound Effects)** — pixabay.com/sound-effects/ — all free, no attribution required. Excellent for UI swooshes, clicks, whooshes.
- **Zapsplat.com** — free with a free account; massive UI/foley library. Great "swoosh", "modal", "click", "chip" sounds. (Free tier asks for attribution or a link; check per file.)
- **Mixkit (mixkit.co/free-sound-effects/)** — free, no attribution, good UI/whoosh/click sounds. Specific tips:
 - **card_deal:** search "card riffle" or "card shuffle" or "dealing cards". Want a crisp, short single deal, not a long shuffle.
 - **chip_push / side_pot_split:** search "poker chips", "casino chips", "chips stack". Grab a shove/clack. For `side_pot_split`, a slightly bigger/longer chip cascade.
 - **modal_swoosh_open/close:** search "ui swoosh", "whoosh soft", "panel open". Pick two that sound related; the close is often just shorter/reversed.
 - **slide_in/slide_out:** search "swoosh transition", "whoosh". Grab ONE good swoosh; you can make `slide_out` by reversing `slide_in` (in Audacity: Effect → Reverse) so they're a true pair.
 - **button_click:** search "ui click subtle", "tick", "soft click". Keep it VERY short/quiet.
 - **frog_toggle:** search "bloop", "water drop", "frog", "boing". Something playful + distinct.

Format for SFX

- Target **.ogg** or **.wav** — CONFIRM which the code expects (your console showed 404s for `.ogg` files: `modal_swoosh_open.ogg`, so the code likely wants **.ogg**). Match that.
- If you download .wav/.mp3, convert to .ogg with **Audacity** (free desktop app): File → Export → Export as OGG. Keep them short and trimmed (no silence at the start — trim dead air so the sound fires instantly).
- Keep each SFX small; total SFX budget <3MB. OGG at moderate quality is fine.

MUSIC (12 tracks — 4 genres x 3 tracks)

Genres: Lounge Jazz, Classical, Bossa Nova, Ambient. The FIRST track of each genre is the free default (must exist); the other 8 are "purchasable" in-app but you still need the files.

Likely folder: `assets/audio/music/<genre>/` e.g.

`assets/audio/music/lounge_jazz/after_hours.mp3`) (your console showed exactly this path). So music = **.mp3**, foldered by genre.

- **Where to get royalty-free music:**
 - **Pixabay Music** (pixabay.com/music/) — free, no attribution, big instrumental library. Search "lounge jazz", "bossa nova", "classical piano", "ambient".
 - **Free Music Archive** (freemusicarchive.org) — CC-licensed; filter for CC0/CC-BY.
 - **Incompetech** (incompetech.com) — Kevin MacLeod's royalty-free library (CC-BY, credit him).
 - **YouTube Audio Library** (studio.youtube.com → Audio Library) — free, downloadable, filter by "no attribution".
- **License note:** prefer CC0 / "no attribution required." If you use CC-BY tracks, you must credit them in the About/Credits modal — keep a note of any track that needs credit.
- **Format/size:** transcode each to **~3MB MP3 (96-128kbps)** in Audacity (Export as MP3). Instrumental only (no lyrics — it's a focus tool). ~2-4 min tracks, they loop/shuffle.
- You need 12 total, but for LAUNCH you really only need the **4 defaults** (one per genre) to exist; the other 8 can be added later. If short on time, get the 4 defaults first.
- Name them simply and confirm the expected names/paths with CC (PART 6). Your console already revealed one: `assets/audio/music/lounge_jazz/after_hours.mp3`.

PART 5 — FILE FORMAT CHEAT SHEET

- **Room backgrounds:** PNG (or JPG — they're full-frame, no transparency needed; JPG saves size, but PNG is safe). 16:9, large (2560x1440+).
- **Everything else visual** (dealer, chips, cards, icons, table, button, tomato): **PNG with transparency**. No exceptions. Draw on transparent canvas, export PNG.
- **SFX:** **.ogg** (your code 404'd on .ogg, so match that), short, trimmed.
- **Music:** **.mp3**, ~3MB, ~96-128kbps, foldered by genre.
- Color profile: sRGB for everything. DPI 72.
- Optimize PNGs before adding (TinyPNG.com or Procreate's export is fine) to keep bundle small.

PART 6 — HOW TO INTEGRATE INTO THE CODE (the safe way)

You do NOT hand-place files into build-wasm/ (that gets wiped). Assets live in the repo's `assets/` folder, which your build copies in. The safe workflow:

Step 1 — FIRST, get the exact filenames and folder paths from Claude Code.

Before you spend hours drawing, open a CC window and ask it (one prompt): "List every visual and audio asset file the code expects: exact filename, exact folder path under assets/, and expected format, for backgrounds, dealer (butler+frog), table, the 52 cards (exact naming scheme), the 8 chips, the dealer button, every UI icon, the tomato icon, side-pot icon, all-in marker, any logo, all 9 SFX, and all 12 music tracks. Grep the asset-loading code (`asset_paths.hpp` / the texture loader / the audio engine) and give me the definitive list. Don't change anything — just report the exact names/paths." This gives you the AUTHORITATIVE names so you name files right the first time. (The names in this guide are best-guesses from the spec; CC's grep is the source of truth.)

Step 2 — Draw/source everything, named exactly per Step 1.

Step 3 — Put them in the repo's assets/ folder.

The build already does `cp -r assets build-wasm/assets`. So the files go in the SOURCE `assets/` directory (e.g. `~/poker-trainer/assets/dealer/butler_profile.png`), matching the paths from Step 1). Drop your files into the matching subfolders there.

Step 4 — Let Claude Code verify placement.

Hand CC a prompt: "I've added the following asset files to assets/: [list]. Verify each is in the correct path the loader expects, the filenames match exactly, and nothing is missing. Report any misnamed/misplaced/missing files. Don't modify the art." This catches typos before you rebuild.

Step 5 — Rebuild + test.

```
cd ~/poker-trainer && source ~/emsdk/emsdk_env.sh && rm -rf build-wasm &&
emcmake cmake -B build-wasm -DCMAKE_BUILD_TYPE=Release && cmake --build
build-wasm -j &&
cp -r assets build-wasm/assets && cd build-wasm && python3 -m http.server 8000
Open localhost:8000 — your art/audio should now appear/play. The 404s you saw before
(after_hours.mp3, modal_swoosh_open.ogg, etc.) should disappear as you fill each file in.
```

Can you do it without CC / messing up?

- Placing files in the right assets/ subfolder and rebuilding is safe and reversible — worst case a file is mis-named and just doesn't load (a 404), which you fix by renaming. You will NOT break the app by adding an asset. So yes, you can do Steps 3+5 yourself.
- The ONE thing to get right is exact filenames/paths — which is why Step 1 (ask CC for the authoritative list) matters. Get that list first and the rest is drag-drop-rebuild.
- You do NOT need to touch any code. Assets are data; the code already knows to load them by name. No code edits required for art/audio (unless you're ADDING new assets the code doesn't reference yet — but you're filling in EXISTING expected assets, so it's pure drop-in).

PART 7 — SMART ORDER TO WORK IN TOMORROW (so you finish)

1. **FIRST (10 min):** run the Step-1 CC prompt, get the authoritative filename/path list. Print it or keep it open. Everything you make gets named from THAT list.
2. **The room background** (draw once, blur 3x). It's the backdrop for everything and one drawing → three files. High value, done early.
3. **The dealer butler** (3 files — your hero asset). Neutral first, derive raised, then profile.
4. **Chips** (draw 1, recolor 8) + **dealer button**. Fast once the first chip's done.
5. **UI icons + tomato** (small, quick, batch them).
6. **Table**.
7. **Cards** (52 — the grind; use a reused template; do this when you can zone out).
8. **Frog** (base + 2 overlays) — lower priority (easter egg); do if time allows.
9. **AUDIO:** grab the 9 SFX (quick — download, trim, convert to .ogg). Then the 4 default music tracks (one per genre) minimum; the other 8 later.
10. **Integrate:** drop everything into assets/ per the paths, run the Step-4 verify prompt, rebuild, test on localhost. Fix any 404s (usually a filename typo).

If you run low on time, MINIMUM viable set for launch:

- 3 backgrounds, 3 butler files, table, 8 chips, dealer button, the UI icons + tomato, 52 cards (functional), the 9 SFX, and the 4 default music tracks. The frog (3) and the 8 paid music tracks can come after launch — the app runs fine without them (they just 404 harmlessly, exactly like now).

PART 8 — QUICK REFERENCE CHECKLIST

VISUAL (transparent PNG unless noted): [] background_root.png (full-frame, heavy blur) [] background_mode.png (full-frame, medium blur) [] background_game.png (full-frame, light blur) [] butler_profile.png (side, faces left) [] butler_neutral.png (front, pass) [] butler_raised.png (front, fail — raised eyebrow/tilted monocle) [] frog_base.png [] frog_expression_pass.png (overlay: blush) [] frog_expression_fail.png (overlay: tongue) [] table.png [] 52 card PNGs (confirm naming) [] chip_white/red/green/black/purple/yellow/brown/gold.png (8) [] dealer_button.png [] icon_shop/help/settings/home/close/exit/copy/share/offline/leaderboard.png [] icon_tomato.png (full-color ok) [] icon_sidepot.png [] icon_allin_marker.png (confirm if needed) [] logo.png (optional — confirm if code wants it)

AUDIO: [] card_deal, button_click, chip_push, side_pot_split, frog_toggle, modal_swoosh_open, modal_swoosh_close, slide_in, slide_out (.ogg, 9 total) [] Lounge Jazz x3, Classical x3, Bossa Nova x3, Ambient x3 (.mp3, 12 total; 4 defaults minimum)

TOOLS: [] Procreate (or Adobe Fresco) for drawing [] Audacity (desktop) for trimming/converting/reversing audio [] Sources: Freesound, Pixabay, Zapsplat, Mixkit (SFX); Pixabay Music, FMA, Incompetech, YT Audio Library (music)

REMEMBER:

- Draw at 2x size, transparent canvas, export PNG.
- Get the authoritative filename list from CC FIRST (Part 6, Step 1).
- Assets go in the repo's assets/ folder; rebuild copies them in. You can't break the app by adding art.
- Match the audio formats the code 404'd on (.ogg for SFX, .mp3 for music).